



Fig. 2: A typical electronic toll collection system (Credit: www.qualix.co.in)

implementing a country-wide solution presents fewer technological risks. In 1959, Nobel Economics Prize winner William Vickrey proposed a system of electronic tolling in which each vehicle was equipped with a transponder.

The transponder's personalised signal was picked up as the vehicle passed through an intersection and relayed to a central computer, which calculated the charge according to the intersection and time of day, and added it to the vehicle's bill.

In 1970s, free flow tolling was successfully tested with vehicles equipped with transponders at under-sides and readers located under the surface of the highways.

Electronic toll collection technology was first introduced in Bergen, Norway in 1986, operating together with traditional toll booths.

The world's first completely unaided full-speed electronic tolling system was introduced in 1991 at Trondheim.

In 1995, Portugal became the first country to apply a single, universal system to all tolls in the country.

The US is known for widespread use of electronic toll collection technology.

In 2005, Santiago, Chile became the world's first city with 100 per cent full-speed electronic tolling with transponders in a system of several urban freeways.

In 2007, United Arab Emirates (UAE) implemented a similar road toll collection in Dubai.

Most modern methods now becoming popular are based on Global Positioning System (GPS). A GPS-based highway toll collec-



Fig. 3: William Vickrey proposed a system of electronic tolling in 1959 (Credit: www.nydailynews.com)

tion system uses a microcontroller (MCU), third-generation (3G) mobile communication connectivity, sophisticated algorithms and complex system integration.

In India, too, electronic toll collection technology is increasingly being made available on toll roads across the country.

Technologies used in an electronic toll collection system

Technologies used in a typical electronic toll collection system include the following major autonomous sub-systems:

Automated vehicle identification (AVI) system. AVI is the method for determining the identity of a vehicle when it is in the toll gate area. Earlier, bar codes were fixed to each vehicle, which were read optically at the toll booth. However, optical systems were found to have poor reading reliability, especially during bad weather conditions and in case of dirty vehicles.

Current AVI systems mostly rely on radio frequency identification (RFID),

where an antenna at the toll gate communicates with a transponder fitted on the vehicle using dedicated short-range communication (DSRC). These have excellent accuracy and can be read at high speeds.

The major disadvantage is the cost of equipping each vehicle with a transponder, which can be a major start-up expense (if paid by the toll agency) or a strong customer deterrent (if paid by the customer).

However, recent models of vehicles are coming factory-fitted with transponders. To preclude the need for transponders, some systems use automatic number plate recognition. Here, a system of cameras captures images of vehicles passing through tolled areas, and images of the number plates are extracted and used to identify the vehicles. This allows customers to use the facility without any advance interaction with the toll agency.

One disadvantage is that the system has a significantly high error rate, giving rise to billing errors. Also, cost of transaction processing (locating and corresponding with customers) can be substantial. Systems based on a manual review stage have lower error rates but, these come with the burden of a continuing staffing expense.

Automated vehicle classification system. Usually, toll facilities have different rates for different types of vehicles. Therefore it becomes necessary to distinguish and classify the vehicles passing through these facilities.

The most common and low-cost method followed is storing vehicle class along with other details in the customer data record. The AVI system identifies a vehicle when it is in the toll gate area, and this AVI data is used to look up vehicle class.

A more complex system uses a variety of sensors. Inductive sensors embedded in the road surface

determine the gaps between vehicles, to provide basic information on the presence of a vehicle. Treadles enable counting the number of axles as a vehicle passes over these. With offset-treadle installations, it can also detect



Fig. 4: Automated vehicle identification system at work (Credit: www.hopelandrfid.com)